

Team HashHawks

CHARKHIA

The wheel that turns crop residue into verified, retirable carbon.

Residue that would otherwise be burned is detected from satellite, routed to a nearby conversion unit, verified in the field, and issued as a carbon credit that can be independently checked and permanently retired. One task identifier threads the entire chain — **detection** → **match** → **evidence** → **decision** → **credential** → **retirement**.

Detection is already a solved problem. We built what happens after it.

ONE AUDITABLE THREAD **Satellite → Soil → Credit**

TEAM

HashHawks · 4 engineers

PROTOCOL

Agent2Agent, open standard

BUILD WINDOW

36–40 hours

LIVE DATA

NASA FIRMS, near-real-time

Burning it is free. Proving you didn't is not.

After the paddy harvest in north-west India, a farmer has a narrow window to clear the field before sowing wheat. Burning the residue costs nothing and takes an afternoon. Every alternative costs money, time, or both.

~28 Mt RICE STUBBLE GENERATED EACH YEAR IN PUNJAB & HARYANA	~80% OF IT BURNED IN THE FIELD, ON COMMON ESTIMATES	~34 Mt CO₂-EQUIVALENT RELEASED BY BURNING ~23 MT OF RESIDUE	40–45% OF DELHI'S PEAK-SEASON AIR POLLUTION, SAFAR ESTIMATE, 2019
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Figures are published estimates and vary by source, season and method. They establish scale, not precision.

The alternative already exists

Crop residue can be pyrolysed into biochar: a stable, carbon-rich solid that locks carbon into soil for centuries rather than releasing it in an afternoon. The conversion technology is mature, and small tractor-mounted units already operate in Punjab and Haryana.

Carbon markets will pay for this. A recognised removal methodology for biochar exists and buyers are actively purchasing against it.

So why is the field still burning?

Because payment depends on **proof**, and proof is the part nobody has industrialised. To issue a credit, someone must establish that a specific quantity of residue was actually collected, actually converted under the right conditions, actually applied to soil — and that the same batch was not already counted by somebody else.

Today that is manual, paper-led, expensive and slow. At a few hundred tonnes it is uneconomic. So the loop never closes, and the cheapest option wins.

THE GAP WE TARGET

The verification gap is the economic gap.

Detection of burning is a solved problem — satellites already see every fire. What does not exist is a cheap, automatic, tamper-evident record of what happened **after** a field was diverted from burning. Charkha builds exactly that, and nothing else.

WHO CARRIES THE COST TODAY

The farmer

Bears the entire cost of the harder choice, and captures none of the carbon value it creates, because nobody can cheaply prove the carbon was sequestered.

WHO CANNOT SCALE

The processor

Can convert residue, but faces manual audit costs that only make sense at volumes small operators will never reach.

WHO CANNOT TRUST

The buyer

Wants removal credits with real permanence, and has no way to check the evidence behind a certificate without taking the issuer's word for it.

We surveyed the field before writing code. Two things are already solved and we do not rebuild them. One thing everyone reaches for, we deliberately avoid. One thing nobody has tried, we use.

<60 min

SATELLITE DETECTION TO OUR DATABASE

Fire detections reach us within an hour of overpass, across MODIS and three VIIRS satellites. Our supply side is seeded by something that actually happened this morning, not by a synthetic fixture.

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GITHUB REPOS · CARBON-CREDITS TOPIC

The top-starred slice is overwhelmingly Solidity token plumbing: ERC-20 and ERC-1155 credits, Algorand blue-carbon MRV, NFT offset marketplaces. Tokenisation is the default reflex here, and the first thing a judge probes.

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AGENT2AGENT × CARBON OR ENERGY

The protocol has 18 repositories under its own organisation and 50+ partner organisations, and nobody has pointed it at this domain. That is the open space — and this problem genuinely needs four independent agents, not one service.

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EVIDENCE CLASSES · PURO CORCCHAR

The real biochar carbon-removal methodology requires production evidence (pyrolysis temperature, residence time, H/C_{org}), application evidence (GPS-tagged location and quantity), and an unbroken chain of custody. Our hash chain **is** the third.

Prior art we studied and deliberately did not repeat

- **Stubble-burn detection is solved.** An existing project already detects crop fires across 23 northern districts using satellite imagery, a fine-tuned vision-language model and live fire data, and exports its model to ONNX. Its architecture stops at detection.
- **Our contribution is everything after detection** — the agent mesh, verified credit issuance, the registry, and the traceability layer.
- **We deliberately avoid blockchain.** A hash chain gives tamper-evidence and a W3C Verifiable Credential gives portable, offline-checkable proof. Consensus buys nothing here and costs a day we do not have.

The live feed almost nobody wires up

- **NASA FIRMS** is a plain REST endpoint with a free access key.
- **5,000 requests per 10 minutes**, MODIS and VIIRS across NOAA-20, NOAA-21 and Suomi-NPP.
- **Detections within 60 minutes** of satellite overpass.
- **Read-only and public.** We pull public data in; nothing of ours goes out.
- **Cached locally**, with a bundled sample behind it, so a dead venue network still produces a populated map.

REFUSED

A token on a chain

Consensus solves disagreement between mutually distrusting writers. Our problem is evidence quality, not write ordering. A hash chain plus a signed credential gives the same trust story in hours instead of days.

REFUSED

Another detector

Satellites already see every fire and published work already classifies them well. Rebuilding that would consume the whole build and add nothing a buyer would pay for.

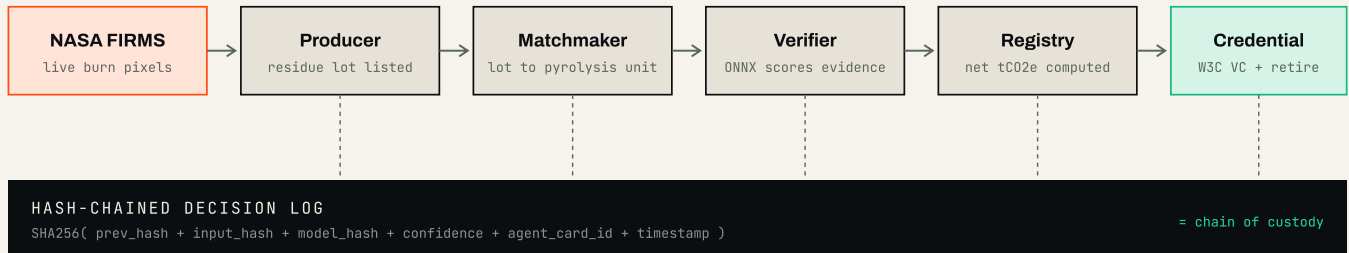
CHOSEN

An open agent protocol

Four independent services with published capability documents. A new producer or processor joins by registering a card, not by us shipping code — which is what makes this a network rather than a demo.

Four agents, one auditable thread.

Each agent is an independent service speaking the open Agent2Agent protocol. Onboarding a new producer or processor is a new Agent Card, not new code. Every decision any agent makes is written once to an append-only, hash-chained ledger.



A2A TASK LIFECYCLE: SUBMITTED → WORKING → INPUT-REQUIRED → COMPLETED — ONE TASK ID THREADS ALL SIX STAGES

How a credit is created

- **Producer** publishes residue lots, seeded from satellite burn detections and registered plots. Detections are deduplicated so re-running ingest cannot fork a lot.
- **Matchmaker** assigns each lot to the accepting conversion unit with the smallest transport debit, inside a radius and within capacity, and records a human-readable rationale for every decision.
- **Verifier** scores field evidence with an ONNX model and validates pyrolysis batch parameters against methodology thresholds.
- **Registry** computes net tonnes of CO₂e from published factors, issues a W3C Verifiable Credential signed with a locally generated key, and retires it through a status list so it cannot be sold twice.

Three interfaces, one origin

- **Operator map.** Live burn detections and conversion units on open map tiles, with matches drawn as routes and a transport-debit summary.
- **Audit console.** The hash chain rendered record by record, re-verified in the browser using the same function the server runs, with a deliberate tamper demonstration.
- **Field capture.** Runs the classifier on the phone itself and submits only a hash and the scores.
- **Public lookup.** Paste any task identifier and follow evidence, decision and credential as a single thread.

What is real, and what is not.

Stating this plainly is a strength, not a weakness. Overclaiming is the fastest way to lose a technical audience.

COMPONENT	STATUS	NOTE
A2A messaging, task lifecycle, Agent Cards	REAL	Built on the official SDK
Hash-chained decision log	REAL	Serialised appends, proven under concurrency
W3C Verifiable Credential issuance	REAL	Locally generated did:key, no registry
ONNX inference, server and browser	REAL	One model file, two runtimes
Live satellite fire feed	REAL	Cached, with a bundled fallback
Message queue (Kafka / NATS)	SIMULATED	In-memory behind a broker-shaped interface
Native mobile ONNX runtime	NOT ATTEMPTED	Deliberately substituted with browser inference
Production auth and infrastructure hardening	PARTIAL	JWT between agents, input validation, rate limiting
Carbon factor certification	CITED, NOT AUDITED	Every constant carries a published source

Data sovereignty is architectural, not a promise

- Field photographs never leave the device.** The classifier runs in the browser; what is submitted is a score, a hash of the image and the model hash. Disconnect the network mid-demonstration and inference still runs.
- No third-party AI services anywhere.** All inference is local, on our own processes and on the user's own device.
- One outbound call, read-only.** A request for public satellite fire detections. Nothing of ours is transmitted.
- Self-hosted end to end.** A single container stack on one machine. No managed service holds our data, and map tiles come from an open source with no vendor account.

Equally plainly, what we do not claim: no end-to-end encryption, no third-party security audit, and no formal regulatory compliance review.

Evidence of rigour.

Every number below is taken from the repository and the continuous integration pipeline, not from a plan.

44 TYPESCRIPT SOURCE FILES	17 AUTOMATED TESTS	12 CI CHECKS PER PUSH	4 PARALLEL WORK STREAMS
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The guarantee we tested hardest

Writing to the decision log is read-then-write: find the tail, hash onto it, insert. Four agents appending at once read the same tail and claim the same sequence number, which silently breaks the chain.

Appends now run inside a transaction holding a database-level advisory lock. To confirm the test guarding this was load-bearing rather than decorative, we removed the lock and re-ran: **fifty concurrent appends failed with a duplicate key on sequence 5**, and the chain stopped verifying. With the lock, all four ledger tests pass.

A check nobody has watched fail is a check you cannot trust.

What continuous integration enforces

- **Correctness:** lint, type check, tests against a real database, production build.
- **Liveness:** all four agents boot and serve a valid Agent Card.
- **Reachability:** a real skill call through the gateway must return a task identifier and a result — added after a transport fault survived a green build.
- **Secrets:** no committed environment file, no private keys, no hard-coded credentials.
- **Data rules:** the build fails if image data appears in a boundary contract or a hosted inference endpoint appears in the source.
- **Deployability:** the container image is built on every change.

How four people work without blocking each other

Shared contracts, the agent wrapper, the ledger and the gateway were written first, so each engineer owns one vertical slice end to end — an agent, its tests, and its interface — and codes against types that already exist. Every change arrives as a reviewed pull request; a red build is not reviewed.

STREAM A Supply side Satellite ingest, residue lots, assignment, operator map.	STREAM B Verification Model training and export, dual-runtime inference, field capture.	STREAM C Credential Carbon mathematics, issuance, retirement, audit console.
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What this becomes outside a weekend.

The architecture was chosen so that the production path is a substitution, not a rewrite. Each item below replaces a component we deliberately scoped down, without changing the contracts between agents.

NEAR**Real message transport and horizontal scale**

The in-memory queue sits behind a broker-shaped interface specifically so it can be replaced with Kafka or NATS without touching agent logic. Agents are already independent processes behind one gateway, so each scales separately.

NEAR**Certified methodology alignment**

Our credential subject is already modelled on the three evidence classes a real biochar removal methodology requires. The next step is formal alignment with a registry standard and independent verifier review, turning a demonstration credit into a sellable one.

MEDIUM**A model trained on real field data**

The pipeline ships before accuracy does, by design. With operator-captured imagery from live sites, the classifier moves from a baseline to a model with measured precision per feedstock class, with active learning driven by every disagreement flagged for review.

MEDIUM**Real logistics, not straight lines**

Greedy assignment over road-adjusted distance is honest and finishes. Production wants genuine road routing, multi-lot consolidation, vehicle capacity and time windows — a scheduling problem, solved against the same interface.

MEDIUM**Portable identity and revocation at scale**

Locally generated keys become managed issuer identities with rotation and published revocation, so a buyer can verify a credit years later without asking us for permission.

LONG**Beyond one crop and one country**

Nothing in the agent contracts is specific to crop residue. The same Producer, Matchmaker, Verifier and Registry pattern covers any waste stream with a conversion pathway and a measurable sequestration claim — and onboarding a new region is a new Agent Card and a new bounding box, not a new system.

Burning it is free. Proving you didn't **isn't**.

A farmer who burns residue pays nothing and is done in an hour. A farmer who diverts it needs someone to collect it, a facility to convert it, and — critically — a credible record that the carbon was actually sequestered. Without that record there is no payment, and without payment there is no reason to choose the harder path.

The verification gap is the economic gap. That is the part we built.

FOR THE GENERATOR

A waste stream becomes a revenue stream, with a receipt they can show anyone.

FOR THE BUYER

A credit whose evidence, model version and decision history can be checked independently, offline.

FOR THE REGULATOR

Tamper-evident custody from satellite detection to retirement, with no trusted intermediary.

PROTOCOL

Agent2Agent, open standard

CREDENTIAL

W3C Verifiable Credentials

INFERENCE

ONNX, local on both ends

CHAIN

SHA-256, append-only